

Multiple Sequence Alignment of Hemoglobin Beta Proteins

CodeAlpha Bioinformatics Internship — Task 2

1. Objective

To perform a multiple sequence alignment of five homologous hemoglobin beta protein sequences using Clustal Omega and identify conserved sequence regions and sequence motifs from the alignment.

2. Introduction

Multiple Sequence Alignment (MSA) is a computational method used to align three or more biological sequences so that corresponding residues can be compared across sequences. Conserved positions can be identified from the alignment and may indicate regions maintained among related proteins. In this task, five hemoglobin beta-family protein sequences from different mammals were aligned using Clustal Omega.

3. Sequences Used

No.	Organism	Protein / UniProt ID	Length shown
1	Mouse	HBB — P02088	147 aa
2	Sheep	HBB — P02075	145 aa
3	Cow	HBB — P02070	145 aa
4	Human	HBB — P68871	147 aa
5	Dog	HBB — P60524	146 aa

4. Methodology

The five protein sequences were obtained from UniProt using accession numbers P68871, P02088, P02075, P60524 and P02070. The sequences were combined in FASTA format and submitted to the EMBL-EBI Clustal Omega multiple sequence alignment service. The resulting alignment was examined using conservation symbols and the colour-coded sequence display.

5. Alignment Result

The alignment was successfully generated by CLUSTAL O (1.2.4). The output contains all five sequences and displays conserved positions beneath the aligned sequences. The colour-coded view provides a visual representation of residue differences and similarities.

Figure 1. Colour-coded Clustal Omega multiple sequence alignment

Coloured sequences

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CLUSTAL O(1.2.4) multiple sequence alignment

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Mouse_HBB_P02088      MVHLTDAEKAASCLWGKVNSEVGGGALGRLLVVPWTQRYFDSFGDLSASAIMGNAK      60
Sheep_HBB_P02075      --MLTAEKKAAVTGFWGKVKVDEVGGGALGRLLVVPWTQRYFEHFGDLSADAVMNNPK      58
Cow_HBB_P02070        --MLTAEKKAAVTAFWGKVKVDEVGGGALGRLLVVPWTQRYFEHFGDLSADAVMNNPK      58
Human_HBB_P68871      MVHLTPEEKSAVTALWGKVNSEVGGGALGRLLVVPWTQRYFEHFGDLSADAVMNNPK      60
Dog_HBB_P60524        -VHLTAEKSLVSGLWGKVNSEVGGGALGRLLVVPWTQRYFEHFGDLSADAVMSNAK      59
                        **  ** : * : ***** : ****.*****:*****:*: *****. .*:*. *
Mouse_HBB_P02088      VKAHGKKVITAFNDGLNHLDSLKGTFASELHCDKLHVDPENFRLGNMIVIVLGHHLG      120
Sheep_HBB_P02075      VKAHGKKVLDSFSNGMKHLDDLKGTFAQLSELHCDKLHVDPENFRLGNVLVVLRHHG      118
Cow_HBB_P02070        VKAHGKKVLDSFSNGMKHLDDLKGTFAALSELHCDKLHVDPENFKLLGNLVVLRNFG      118
Human_HBB_P68871      VKAHGKKVLGAFSDGLAHLNKLKGTFAQLSELHCDKLHVDPENFRLGNVLCVLAHHFG      120
Dog_HBB_P60524        VKAHGKKVLSFSDDLKHLNKLKGTFAKLSELHCDKLHVDPENFKLLGNVLCVLAHHFG      119
                        *****: :*.:* :*.***** *****:*****:*****:*: **.: : *
Mouse_HBB_P02088      KDFTPAQAQAFQKVVAGVATALAHKYH      147
Sheep_HBB_P02075      NEFTPVLAQDFQKVVAGVANALAHKYH      145
Cow_HBB_P02070        KEFTPVLAQDFQKVVAGVANALAHKYH      145
Human_HBB_P68871      KEFTPPVQAAYQKVVAGVANALAHKYH      147
Dog_HBB_P60524        KEFTPVQAAYQKVVAGVANALAHKYH      146
                        : :***  ** :*****.*****:**
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Source: Clustal Omega alignment generated during the analysis.

6. Interpretation of Conservation

In the Clustal output, an asterisk (*) marks positions where the same residue is present in all aligned sequences. A colon (:) indicates a strongly conserved substitution, while a period (.) indicates a weaker similarity according to the alignment convention.

The alignment shows strongly conserved blocks around the sequence beginning with **VKAHGKKV**, the central region containing **HCDKLHVDPENFRLG**, and the C-terminal region ending around **VAGANALAHKYH**. These are direct observations from the conservation symbols in the generated alignment.

7. Conserved Sequence Regions / Motifs

Observed region	Observation
VKAHGKKV...	Strong conservation across the five aligned sequences
...HCDKLHVDPENFRLG...	Highly conserved central region
...VAGANALAHKYH	Highly conserved C-terminal region

8. Discussion

The five hemoglobin beta-family sequences are similar in overall sequence but differ at several positions and in total length. The many identical positions demonstrate substantial conservation within this protein family. The conserved blocks identified above are suitable regions for reporting sequence conservation. The alignment alone does not establish a specific biological function for each conserved region; functional interpretation would require supporting annotations or literature.

9. Conclusion

A successful multiple sequence alignment was generated for five hemoglobin beta proteins from mouse, sheep, cow, human and dog using Clustal Omega. The alignment revealed numerous conserved positions and strongly conserved sequence regions, including regions around VKAHGKKV, HCDKLHVDPENFRLG and the C-terminal VAGANALAHKYH sequence. This demonstrates how MSA can be used to compare related protein sequences and identify conserved regions.

10. References

- UniProt Knowledgebase — protein sequence records for P68871, P02088, P02075, P60524 and P02070.
- EMBL-EBI Clustal Omega — multiple sequence alignment service.
- CodeAlpha Bioinformatics Internship — Task 2 instructions.

Note: Sequence lengths and conservation observations are taken from the alignment generated during this Task 2 analysis.